A New Master-Slave Based Secondary Control Method for Virtual Synchronous Generator

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***Abstract*—VSG (Virtual Synchronous Generator) is a control method for inverters that mimics both the steady-state droop characteristic and transient characteristic of Synchronous Generators (SGs). Because the steady states of VSG and droop controlled inverters are similar, the deviation of frequency and voltage amplitude also exists in the VSG. This paper proposes a new secondary control method mimicking the master-slave control but requiring no communications among VSGs. In the proposed method, the master VSG is controlled as a voltage source while the slave VSGs are controlled as current sources. The droop characteristic biases of slave VSGs are regulated online based on the load power estimation. Through this method, frequency and voltage deviation can be compensated and the majority of load power can be shared equally among slave VSGs. Simulation and experimental results are provided to verify the effectiveness of the proposed control strategy.**

Keywords—VSG; master-slave control; secondary control

I. INTRODUCTION

Microgrid is a newly developed concept emerged with the increasing concern about the energy crisis and environmental problems. It is an integrated energy system consisting of DGs (Distributed Generators) and interconnected loads. Although the DGs include different kinds of renewable-energy sources such as wind power, photovoltaic and fuel cell etc, they are usually connected to a common bus through power electronic interfaces like inverters. The control method for the inverters is an important part of a microgrid. One control method is called a droop control. The traditional droop method mimics the droop characteristics of SGs, allowing load power sharing among parallel inverters by regulating the amplitude and frequency of inverter output voltages according to their output active and reactive power. Compared with the conventional bulk power plants, in which SGs dominate, the DG units have either small or no rotor inertia and damping effects [2]. With the increase in the use of DGs, the negative impact of low inertia on the system stability and dynamic performance will increase. Therefore, the virtual synchronous generator(VSG) method has been proposed as a way to improve performance. This method enhances the system frequency stability by mimicking the steady-state droop characteristics and the dynamic characteristics of SGs[2-13]. However, the steady-state characteristic of VSG and that of droop controlled inverters are similar. Therefore, the deviation of frequency and voltage amplitude also exists in VSG. The frequency and voltage deviation is due to the inherent tradeoff in the droop characteristic between the power sharing accuracy and the voltage regulation rate.

To cope with this problem, a method for sharing reactive power proportional to VSGs' capacity and compensating the voltage deviation was proposed in [14]. However, each paralleled VSG has to know the voltage amplitude and output power of other VSGs, which requires a mass of communication lines. The method proposed in [15] uses a central controller to send commands to all the parallel VSGs to compensate frequency and voltage deviation synchronously. This method can compensate the frequency and voltage deviation effectively. However, the reliability of the system is low since the fail down of the central controller would lead to the collapse of the whole system. A secondary method based on load power estimation for droop controlled inverters to realize the frequency and voltage compensation and power sharing was proposed in paper [16]. The master inverter adopts conventional droop method while the slave inverters adopt reversed droop method.

This paper proposes a new secondary control method mimicking the master-slave control but requiring no communications among VSGs. In the proposed method, the master VSG is controlled as a voltage source[11] while the slave VSGs are controlled as current sources[6]. This new secondary control method for VSG can realize the restoration of frequency and voltage amplitude. The master VSG can work at nominal power rate and the majority of load power demand can be equally shared among slave VSGs.

II. REVIEW ABOUT VIRTUAL SYNCHRONOUS GENERATOR

In this paper, the current source controlled VSG [6] and voltage source controlled VSG [11] are adopted as slave VSGs and master VSG. The introduction of these two method is as follows.

A. VSG(KU Leuven)[6]

The swing equation of SGs is shown as follows.

$$P_m - P_e = J\omega\frac{d\omega}{dt} \tag{1}$$

This work was supported by the National Natural Science Foundation of China under Grant 51437007, and the Power Electronics Science and Education Development Program of Delta Environmental & Educational Foundation under Grant DREM2014002.

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which equals to

$$\frac{P_m}{\omega}-\frac{P_e}{\omega}=T_m-T_e=J\frac{d\omega}{dt} \tag{2}$$

where T_m and T_e are the mechanical and electrical torque, P_m and P_e are the mechanical and electrical power. ω is the grid angular frequency. J is the moment of inertia. When the load power increases (namely P_e increases), P_m will be smaller than P_e, the rotating speed of rotor will decrease and the kinetic energy of rotor will be released to make up this power deficiency and vice versa. So the output active power of rotor is $-J\omega\frac{d\omega}{dt}$.

To mimic the rotor inertia, the author defines the virtual inertia power command:

$$P_{inertia}=-J\omega\frac{d\omega}{dt} \tag{3}$$

When the grid angular frequency reaches steady state the virtual inertia power command will equal to zero namely this term only functions in the transient process.

To mimic the primary frequency regulation of SGs, the author defines the power command based on the droop characteristic,

$$P_{droop}=-\frac{1}{m}\left(\omega-\omega_0^{'}\right) \tag{4}$$

where $1/m$ is the droop coefficient and $\omega_0^{'}$ is the frequency when the dispatched power is zero.

To realize the primary frequency regulation and mimic the rotor inertia simultaneously. the output active power commend of inverter is the combination of these two terms,

$$P_{ref}=P_{inertia}+P_{droop}=-J\omega\frac{d\omega}{dt}-\frac{1}{m}\left(\omega-\omega_{ref}\right) \tag{5}$$

According to the instantaneous power theory, the active power and reactive power equal to

$$\begin{cases} p=\dfrac{3}{2}\left(u_d i_d+u_q i_q\right) \\ q=\dfrac{3}{2}\left(u_d i_q-u_q i_d\right) \end{cases} \tag{6}$$

u_q=0 when the control of inverter is based on the grid voltage, and the equation can be simplified as,

$$\begin{cases} p=\dfrac{3}{2}u_d i_d \\ q=\dfrac{3}{2}u_d i_q \end{cases} \tag{7}$$

Therefore, the active and reactive current references when the inverters operate in the unity power factor are as follows:

$$\begin{cases} i_d^{\,*}=\dfrac{2P_{ref}}{3u_d} \\ i_q^{\,*}=0 \end{cases} \tag{8}$$

Then the current references are feed to the inner current loop to produce the PWM switch signal. The control block is shown in Fig.1.

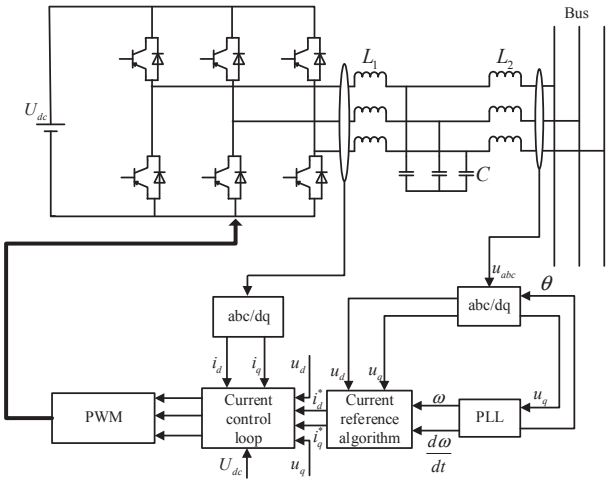


Fig. 1. Main circuit and control block of VSG method

B. ISE lab's VSG method[11]

The control block is shown as Fig.2. In this method, the well-known swing equation of SGs is the heart of VSG control.

$$P_{in}-P_{out}-D\left(\omega_m-\omega_g\right)=J\omega_m\frac{d\omega_m}{dt} \tag{9}$$

where P_{in} is the virtual mechanical power determined by the

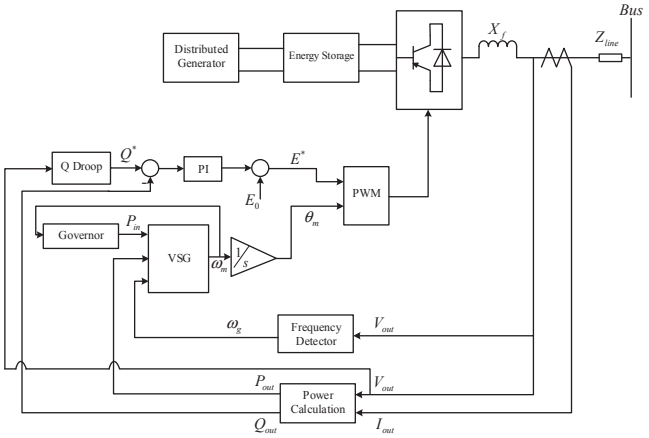


Fig. 2. The block diagram of the ISE's VSG control method

governor, P_{out} is the measured output power, J is the virtual inertia, D is the virtual damping factor, ω_m is the virtual rotor angular frequency, ω_g is the grid angular frequency.

The block "Governor" in Fig.2 is a ω–P droop controller which can be represented as

$$P_{in}=P_0-k_p(\omega_m-\omega_0) \tag{10}$$

P_0 is the set value of active power, k_p is the droop coefficient, ω_0 is the nominal angular frequency.

The block "Q droop" is a V-Q droop controller which can be represented as

$$Q_{ref}=Q_0-k_q\left(V_{out}-V_0\right) \tag{11}$$

Q_0 is the set value of reactive power, k_q is the droop coefficient for reactive power control, V_{out} is the output voltage amplitude,

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V_0 is the set value of voltage amplitude, Then the voltage command produces the PWM switching signal.

III. Proposed Control Method

Take a N-parallel-VSGs system for example. The main VSG adopts the ISE lab's VSG method and slave VSGs adopt the current source controlled VSG. If slave VSGs don't operate in the unity power factor, the real and reactive power reference are as follows.

$$P_n^* = P_{inertia} + P_{droop} = -J\omega\frac{d\omega}{dt} - k_{pn}^{'}\left(\omega - \omega_0^{'}\right) \quad (12)$$

$$Q_n^* = -k_{qn}^{'}\left(E - E_0^{'}\right) \quad (13)$$

where P_n^* and Q_n^* are generated reference of active and reactive power, $\omega_0^{'}$ and E_0' are frequency and voltage amplitude when the dispatched power is zero. k_{pn}' and k_{qn}' are the droop coefficient of the nth slave VSG, J is the moment of inertia. The droop coefficients of slave VSGs are designed according to the relation in (14) and (15).

$$k_{p1}^{'} : k_{p2}^{'} : \ldots : k_{p(\mathrm{N}-1)}^{'} = P_{r1} : P_{r2} : \ldots : P_{r(\mathrm{N}-1)} \quad (14)$$

$$k_{q1}^{'} : k_{q2}^{'} : \ldots : k_{\mathrm{q}(\mathrm{N}-1)}^{'} = Q_{r1} : Q_{r2} : \ldots : Q_{r(\mathrm{N}-1)} \quad (15)$$

where P_{rn} and Q_{rn} are respectively the real and reactive power rating of nth slave VSG.

Combining the swing equation and governor equation together and linearizing it, the real power control for master VSG is shown as follows.

$$\omega_m = \omega_0 - \frac{1}{k_p + D + J\omega_0 s}\left(P - P_0\right) \quad (16)$$

The reactive power control loop is as follows.

$$E = E_0 - k_q(Q - Q_0) \quad (17)$$

where P_0 and Q_0 are set value of real and reactive power, P and Q are the output real and reactive power, k_p and k_q are droop coefficient, ω_0 and E_0 are the nominal angular frequency and voltage amplitude, D is the damping effect term, J is the rotor inertia.

The whole process contains following steps.

1) The PLL block in slave VSG detects the real time frequency and voltage amplitude to make sure whether they are within the normal operation range, that's to say: $\omega_c \in [\omega_L, \omega_H]$, $E_c \in [E_L, E_H]$,where ω_L and ω_H are respectively the floor and the ceiling value of frequency. E_L and E_H are respectively the floor and the ceiling value of voltage amplitude. When ω_c or E_c exceeds the normal operation range, the secondary control will be switched on.

2) When the secondary control is triggered, the slave VSG will stand by for a short period T_{d1} to make sure the system has reached the steady state. Then the slave VSG will estimate the load power and calculate the new droop bias. Another delay T_{d2} is added intentionally before the change of the droop characteristic in consideration of the different calculation speed of different slave VSGs. Then the slave VSGs stop detecting the real time frequency and voltage amplitude.

3) The slave VSGs adjust their own droop characteristics based on the new droop bias.

4) The slave VSGs stand by for a short period T_{d3} to make sure the system has reached the steady state again. Then they restart detecting the real time frequency and voltage amplitude as in step 1).

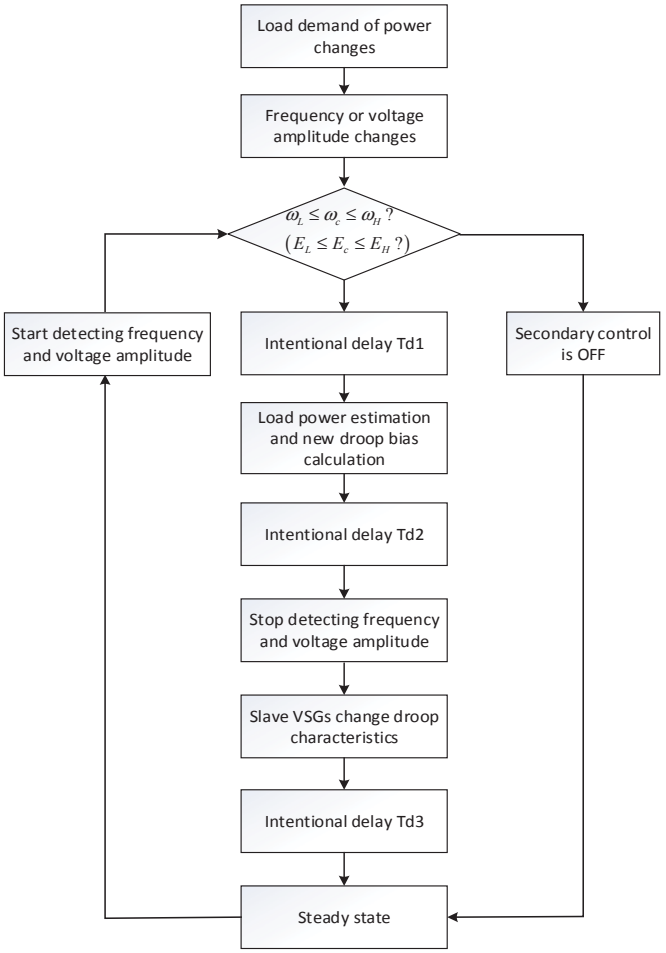


Fig. 3. Flow chart of the proposed method

The calculations of load power and droop bias in slave VSGs in step 2) are as follows.

(1) Calculating the output real and reactive power of nth slave VSG and master VSG.

$$\begin{cases} P_{cn} = -k_{pn}^{'}(\omega_c - \omega_0^{'}) \\ Q_{cn} = -k_{qn}^{'}(E_c - E_0^{'}) \end{cases} \quad (18)$$

$$\begin{cases} P_c = P_0 - \left(k_p + D\right)\left(\omega_c - \omega_0\right) \\ Q_c = Q_0 - \left(E_c - E_0\right)/k_q \end{cases} \quad (19)$$

where ω_c and E_c are the detecting real time frequency and voltage amplitude, P_{cn} and Q_{cn} are the output real and reactive power of nth slave VSG. P_c and Q_c are the output real and reactive power of master VSG.

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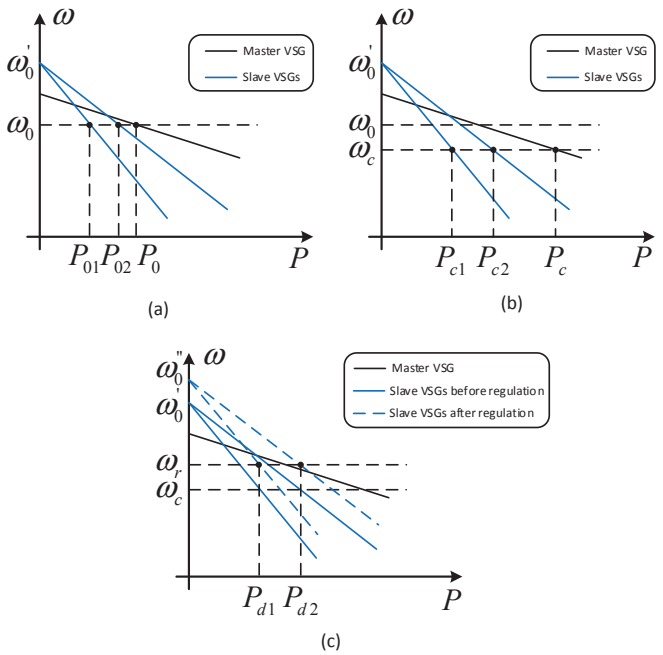


Fig. 4. System status under different conditions: (a) before load changing. (b) load changes but secondary control hasn't been triggered. (c)secondary ontrol is triggered.

(2) Estimating the total load real and reactive power demand P_t and Q_t as follows.

$$P_t = P_c + P_{c1} + P_{c2} + \cdots + P_{c(N-1)} \tag{20}$$

$$Q_t = Q_c + Q_{c1} + Q_{c2} + \cdots + Q_{c(N-1)} \tag{21}$$

(3) Calculating the output power P_r and Q_r of master VSG after the secondary control, namely the output power at nominal frequency ω_r and voltage amplitude E_r.

$$P_r = P_0 - \left(k_p + D\right)\left(\omega_r - \omega_0\right) \tag{22}$$

$$Q_r = Q_0 + \left(E_0 - E_r\right)/k_q \tag{23}$$

(4) Calculating the real and reactive power demand P_{dn} and Q_{dn} for nth slave VSG.

$$P_{dn} = \frac{k'_{pn}}{k'_{p1} + k'_{p2} + \cdots + k'_{p(N-1)}} \cdot \left(P_t - P_r\right) \tag{24}$$

$$Q_{dn} = \frac{k'_{qn}}{k'_{q1} + k'_{q2} + \cdots + k'_{q(N-1)}} \cdot \left(Q_t - Q_r\right) \tag{25}$$

(5) The new bias of the droop characteristic for the nth slave VSG based on (12), (13), (24) and (25) are as follows.

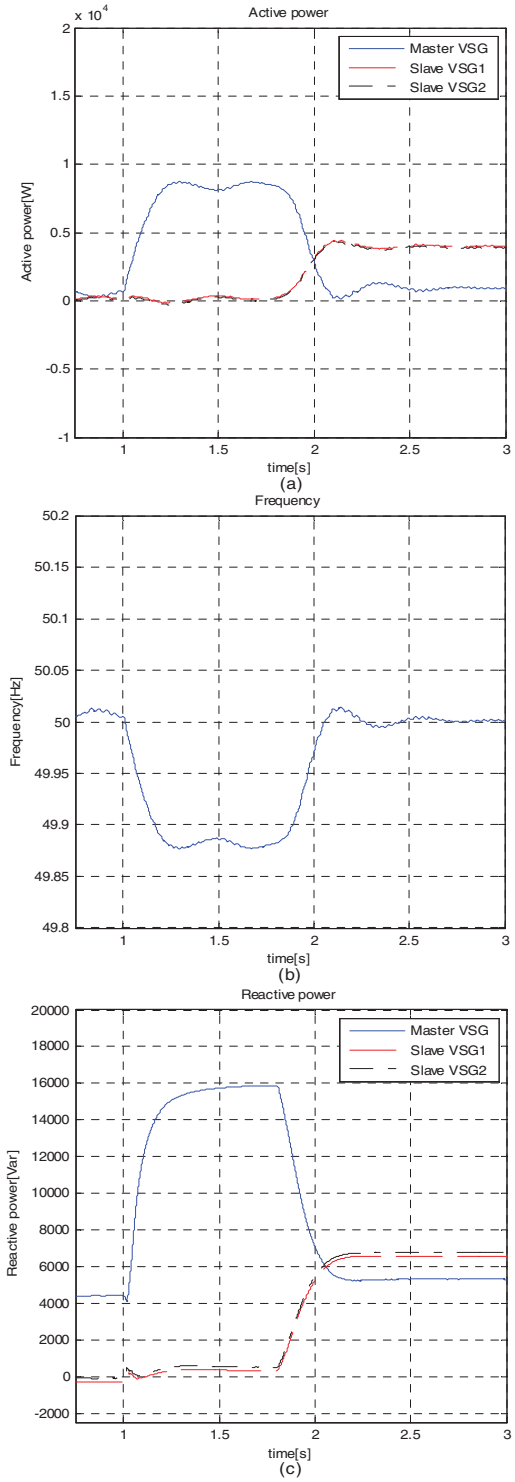
$$P_n^* = -k'_{pn}\left[\omega - \left(\omega_r + \frac{P_{dn}}{k'_{pn}}\right)\right] \tag{26}$$

$$Q_n^* = -k'_{qn}\left[E - \left(E_r + \frac{Q_{dn}}{k'_{qn}}\right)\right] \tag{27}$$

where $\omega_r + P_{dn}/k'_{pn}$ can be regarded as the new $\omega_0^{'}$ after the regulation. Fig.3 shows the flow chart of the regulation process, which is also schematically illustrated in Fig.4.

IV. Simulation And Experiment Results

PSCAD simulations are conducted to verify the proposed method and the waveforms are drawn in MATLAB. The simulation system includes one master VSG and two slave VSGs. The load active power increases at 1s, of which the master VSG takes charge of the majority. The system frequency drops from 50.01Hz to 49.87Hz. The secondary control for frequency starts manually at 1.8s.



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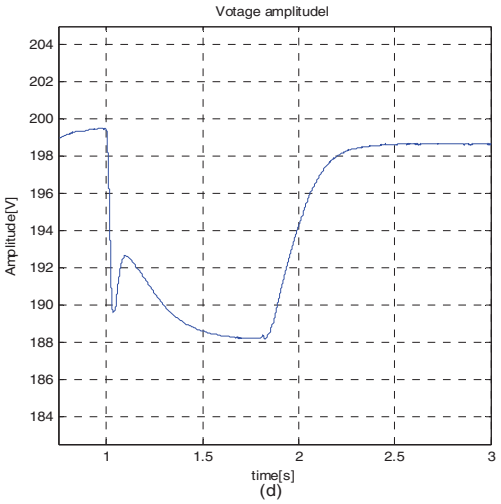


Fig. 5. Simulation waveforms of (a)active power (b) frequency (c)reactive power (d)voltage amplitude

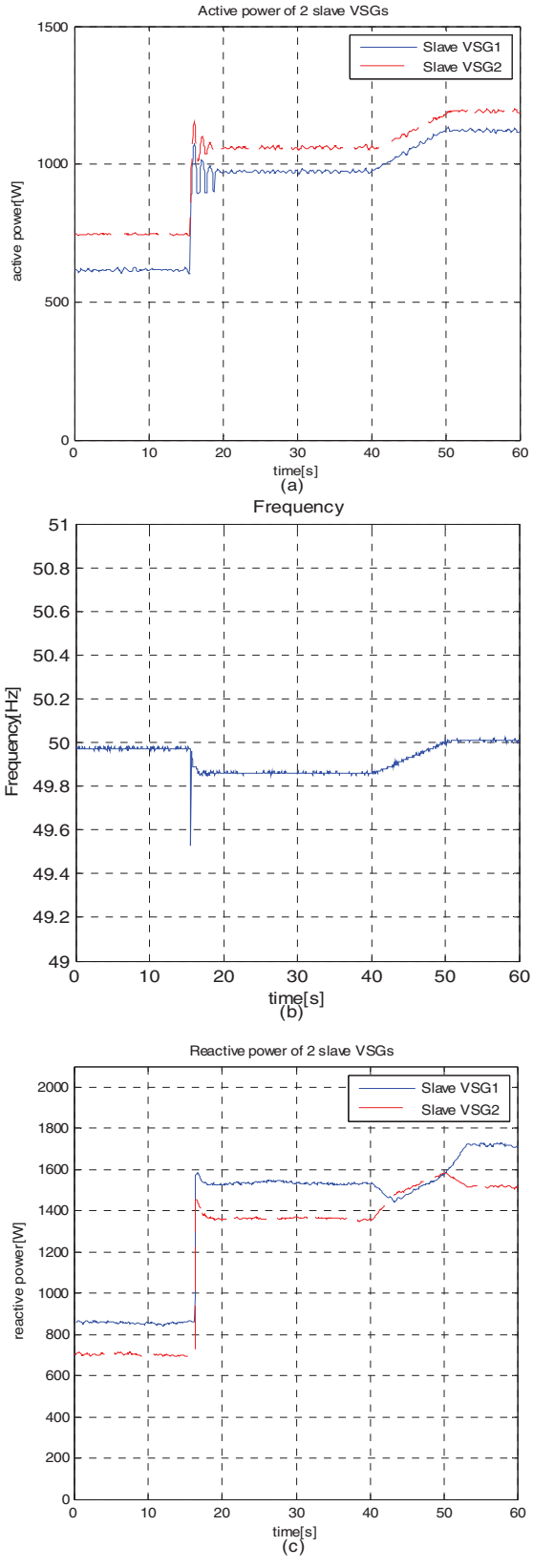
We can see that the output active power of slave VSGs increases while the output power of master VSG decreases in Fig.5(a). The increment of load active power demand is equally shared among the slave VSGs. The system frequency is restored to the rated value ωr in Fig.5(b).

The load reactive power increases at 1s, of which the master VSG takes charge of the majority. The system voltage amplitude drops from 199.5V to 188V. The secondary control for voltage amplitude starts manually at 1.8s. We can see that the output reactive power of slave VSGs increases while the output power of master VSG decreases in Fig.5(c). The increment of load reactive power demand is equally shared among the slave VSGs. The system voltage amplitude is restored to the rated value E_r in Fig.5(d).

The hardware experiments are conducted in a platform consisting of three similar parallel three-phase inverters (MYINV-9R144). One is controlled as the master and the other two as the slaves. The waveforms of the output power, the frequency and the voltage amplitude are drawn in MATLAB based on the recorded data by the HIOKI Power Analyzer 3390. In the first experiment, the load active power increase at 15s and the frequency secondary control is triggered at 40s as shown in Fig.6(a). It is easy to find that the frequency drops from 49.97 Hz to 49.87 Hz at 15s and it recovers to 50 Hz at 40s in Fig.6(b). The output active power of 2 slave VSGs increase at 40s. In the second experiment, the load reactive power increase at 15s and the voltage secondary control is triggered at 40s. The results are similar.

V. Conclusion

This paper proposes a new secondary control method mimicking the master-slave control but requiring no communications among VSGs. In the proposed method, the master VSG is controlled as a voltage source while the slave VSGs are controlled as current sources. This new secondary control method for VSG can realize the restoration of frequency and voltage amplitude. The master VSG can work at nominal power rate and the majority of load power demand can be equally shared among slave VSGs.



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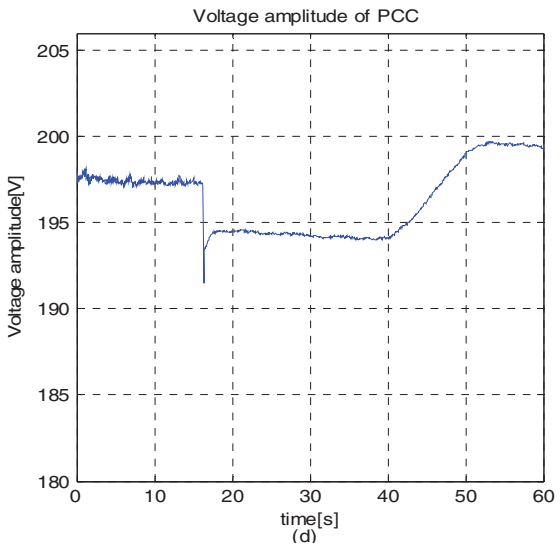


Fig. 6. Experimental waveforms of (a)active power (b) frequency (c)reactive power (d)voltage amplitude

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